

SUMMER RESEARCH INTERNSHIP PROGRAMME

2026

FINAL RESEARCH REPORT — SUBMISSION COVER PAGE

MedSAM-U: Uncertainty-Guided Auto Multi-Prompt Adaptation for Reliable MedSAM

PROGRAMME DETAILS

Domain / Track:	Medical Image Segmentation / Deep Learning	Duration:	2026 Research Internship
Department:	Artificial Intelligence & Machine Learning	Batch / Year:	B.Tech / 6th Semester

SUBMITTED BY

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UNDER THE MENTORSHIP OF

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DECLARATION

We hereby declare that this report is a record of genuine work carried out by us under the guidance of the above-mentioned mentor, as part of the IEEE SMC Student Branch Chapter, KGEC — Research Internship Programme 2026, and has not been submitted elsewhere for any other purpose.

Mentor's Signature

Programme Coordinator

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1. Introduction

Medical image segmentation involves identifying and separating the part of an image that is of interest from the surrounding region. It is commonly used to locate lesions, organs, and other anatomical structures in medical images such as CT, MRI, ultrasound, dermoscopy, and colonoscopy. Foundation models have made prompt-based segmentation more practical, although the quality of the segmentation can still depend on how accurately the target is prompted.

SAM introduced a general framework in which segmentation can be guided using prompts. MedSAM applies the same idea to medical images, making it more suitable for medical image segmentation tasks [2].

MedSAM-U builds on MedSAM and focuses on a practical issue: the predicted mask can change when the position or type of the input prompt changes [1].

1.1 Problem Statement

In a typical MedSAM workflow, a bounding box is used to roughly indicate where the target is located. It does not have to match the target perfectly, but it should cover it reasonably well. If the box is poorly placed, the resulting segmentation can also be affected. MedSAM-U explores whether the differences between predictions from several slightly different prompts can be used to find uncertain regions and guide a better prompt.

1.2 Objective

- Understand the MedSAM-U framework and the roles of its three main components: MPA-MedSAM, UGMP, and UGPA.
 - Study how different prompt variations can be used to estimate uncertainty.
 - Implement the UGMP stage using the MedSAM inference pipeline in Google Colab.
 - Generate multiple segmentation masks and probability maps, and use them to create an entropy-based uncertainty map.
- Understand how uncertainty-guided prompting is different from the usual single-prompt MedSAM workflow.**

1.3 Scope of This Work

The research paper presents the complete MedSAM-U framework. The implementation completed during this internship focuses on the UGMP stage. It is treated as a prototype for understanding the uncertainty-estimation mechanism rather than as a complete reproduction of the paper's training and benchmark experiments.

2. Background and Proposed Framework

2.1 SAM and MedSAM

SAM is a promptable segmentation foundation model that can use inputs such as points and bounding boxes. MedSAM adapts the segmentation framework to medical images. Its purpose is to make a general segmentation model more useful across different medical targets and imaging modalities [2].

2.2 MedSAM-U

MedSAM-U consists of three connected parts. MPA-MedSAM adapts the model for multi-prompt use. UGMP, or Uncertainty-Guided Multi-Prompt Generation, produces several predictions from slightly different prompts and estimates their uncertainty. UGPA, or Uncertainty-Guided Prompt Adaptation, uses uncertain regions to construct improved prompts and performs segmentation again [1].

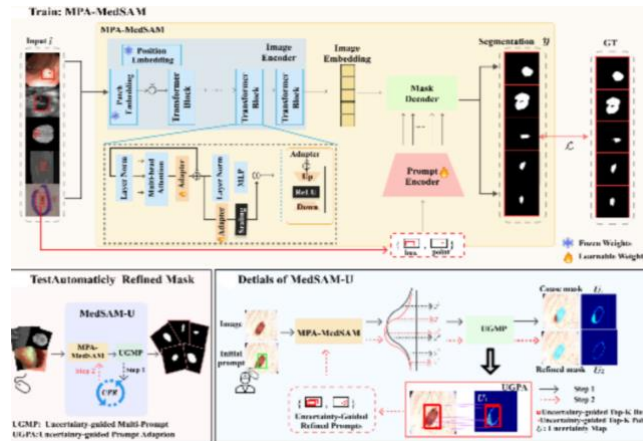


Fig. 1. Overview of the MedSAM-U framework. Adapted from [1].

2.3 MPA-MedSAM

MPA-MedSAM introduces adapter modules into the image encoder and allows the prompt encoder to be trained for the adapted setting. The resulting model can work with both point-based and box-based prompts. This provides the multi-prompt capability needed by the later uncertainty-guided stages [1].

2.4 UGMP

UGMP starts with an initial bounding box and generates several nearby versions of it. Each prompt is processed separately to obtain a segmentation prediction. When the predictions remain similar, the model is relatively stable in that region. When they differ, the region is treated as more uncertain.

2.5 UGPA

UGPA uses the uncertainty information generated by UGMP. Regions with high uncertainty are used to construct refined bounding boxes and point prompts. MedSAM is then run again using the refined prompts. The purpose is to reduce uncertainty while improving the final segmentation [1].

3. Methodology and Implementation

3.1 UGMP Implementation Pipeline

The implementation was carried out in Google Colab using the MedSAM repository and its inference pipeline. The UGMP process was implemented step by step. It started with a reference bounding box, followed by generating several slightly modified versions of the same box. Each of these prompts was then given to MedSAM to obtain a segmentation. The probability maps from these predictions were saved and used to calculate the entropy-based uncertainty.

- An initial bounding box was selected on the demo medical image.
- Five nearby bounding boxes were generated by making small random changes to the box coordinates.
- Each perturbed box was passed separately through MedSAM.
- The resulting binary masks were stored for visualization.
- The continuous probability maps were retained for uncertainty estimation.
- Pixel-wise entropy was calculated from the predictions to produce the uncertainty map.

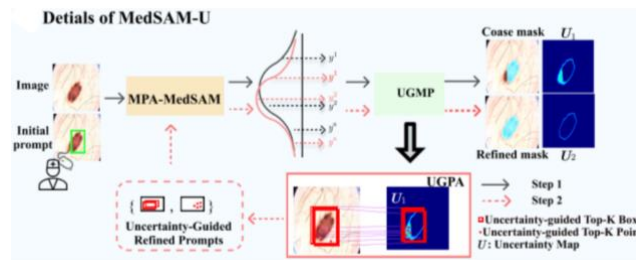


Fig. 2. UGMP process and prompt perturbation used in the MedSAM-U methodology [1].

3.2 Bounding Boxes and Masks

A bounding box is a rectangular prompt defined by four coordinates, usually representing the left, top, right, and bottom boundaries. It gives MedSAM an approximate location of the target. The mask is the pixel-level segmentation generated by the model. In this implementation, the box was changed slightly several times, and the resulting masks were compared to see how stable the segmentation remained.

3.3 Entropy-Based Uncertainty

For a binary prediction with probability p , the entropy is given by $H(p) = -p \log(p) - (1-p) \log(1-p)$. Entropy is low when the prediction is close to 0 or 1, while it becomes higher when the prediction is less certain. In UGMP, the probability maps from multiple prompt-conditioned predictions are used to identify pixels where the model's predictions differ [1].

4. Results and Discussion

4.1 Implementation Results

The five perturbed bounding boxes resulted in five segmentation masks. Most of the masks were similar in the central part of the target, while small changes could be seen around the boundary. This showed the main purpose of the prototype: the multiple masks are useful not just as separate predictions, but as a way of measuring how uncertain the model is.

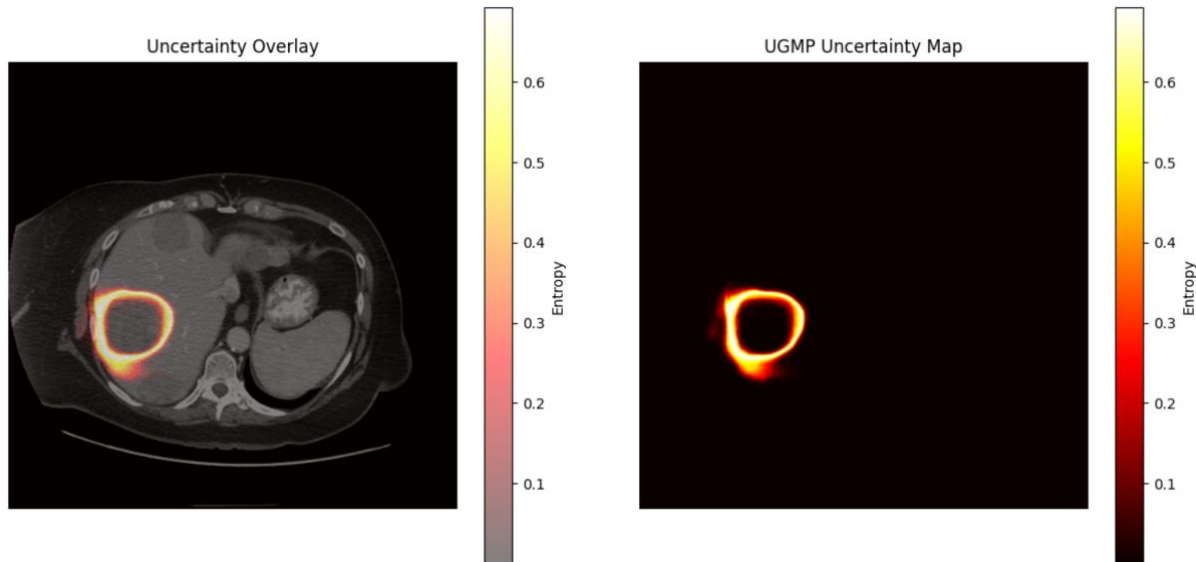


Fig. 3. Uncertainty visualizations from the UGMP prototype. Left: uncertainty overlay. Right: entropy map.

The brighter areas are concentrated mainly around the target boundary. This is what we would expect from the method, since the central part of the target tends to remain stable while small changes in the prompt can cause the predicted boundary to move.

4.2 Paper Results

The paper evaluates MedSAM-U on five datasets representing different medical imaging modalities: ISIC-2017 for dermoscopy, Kvasir-SEG for colonoscopy, CT2US for ultrasound, KiTS23 for CT, and BraTS 2021 for MRI [1].

Method	Average Dice
SAM	72.8%
MedSAM	84.4%
MedSAM-U	86.1%

Table 1. Reported average Dice scores from the MedSAM-U paper [1].

The reported average Dice Score improves from 72.8% with SAM to 84.4% with MedSAM and then to 86.1% with MedSAM-U. The paper also reports improvements of up to 20.5% when the starting prompts are of low quality. These results support the idea that uncertainty can be useful when the initial prompt is not sufficiently reliable [1].

4.3 Interpretation

The key difference from standard single-prompt MedSAM inference is the use of uncertainty as feedback. Rather than using one bounding box and stopping after a single prediction, MedSAM-U checks how the segmentation changes when nearby prompts are used. The differences between these outputs then indicate where further refinement may be needed.

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5. Discussion

5.1 Strengths

- The method reduces dependence on a perfectly accurate initial prompt.
- Multiple predictions provide a direct way to observe prompt sensitivity.
- The UGMP uncertainty estimate does not require a separate uncertainty network.
- The paper evaluates the approach across several medical imaging modalities.

5.2 Limitations

- Several prompt-conditioned inferences increase computational cost.
- The reported experiments mainly focus on 2D medical images.
- The method still starts from an initial prompt that roughly identifies the target.
- The internship implementation covers UGMP rather than the complete MPA-MedSAM and UGPA benchmark pipeline.

5.3 Future Work

The paper suggests making uncertainty estimation more closely integrated with the adapter modules and moving towards a more automatic medical image annotation system [1]. For the implementation carried out during this internship, the next practical step would be to reproduce UGPA and then test the complete pipeline on one of the benchmark datasets.

6. Conclusion

MedSAM-U focuses on a practical limitation of prompt-based medical segmentation: the output can change when the initial prompt is inaccurate. The framework brings together MPA-MedSAM, UGMP, and UGPA so that the differences between multiple predictions can be used to guide prompt refinement.

The UGMP prototype implemented in Google Colab demonstrated the basic uncertainty-estimation process. Five slightly perturbed bounding boxes were generated, and MedSAM produced a prediction for each one. Their probability maps were then used to calculate an entropy-based uncertainty map. In the resulting visualization, uncertainty was concentrated mainly around the target boundary.

Working through the implementation gave practical experience with prompt perturbation, mask generation, probability maps, and entropy-based uncertainty estimation. It also helped maintain a clear distinction between the complete MedSAM-U framework described in the paper and the UGMP portion reproduced during the internship.

7. AI Usage Declaration

AI-based tools were used mainly to understand the MedSAM-U pipeline and to get a clearer understanding of technical concepts related to medical image segmentation, prompt generation, and uncertainty estimation.

8. References

- [1] N. Zhou, K. Zou, K. Ren, M. Luo, L. He, M. Wang, Y. Chen, Y. Zhang, H. Chen, and H. Fu, “MedSAM-U: Uncertainty-Guided Auto Multi-Prompt Adaptation for Reliable MedSAM,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 36, no. 3, pp. 3768–3781, Mar. 2026, doi: 10.1109/TCSVT.2025.3620396.
- [2] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, “Segment Anything in Medical Images,” *Nature Communications*, vol. 15, Art. no. 654, 2024.
- [3] B. Wang Lab, “MedSAM: Segment Anything in Medical Images,” GitHub repository.